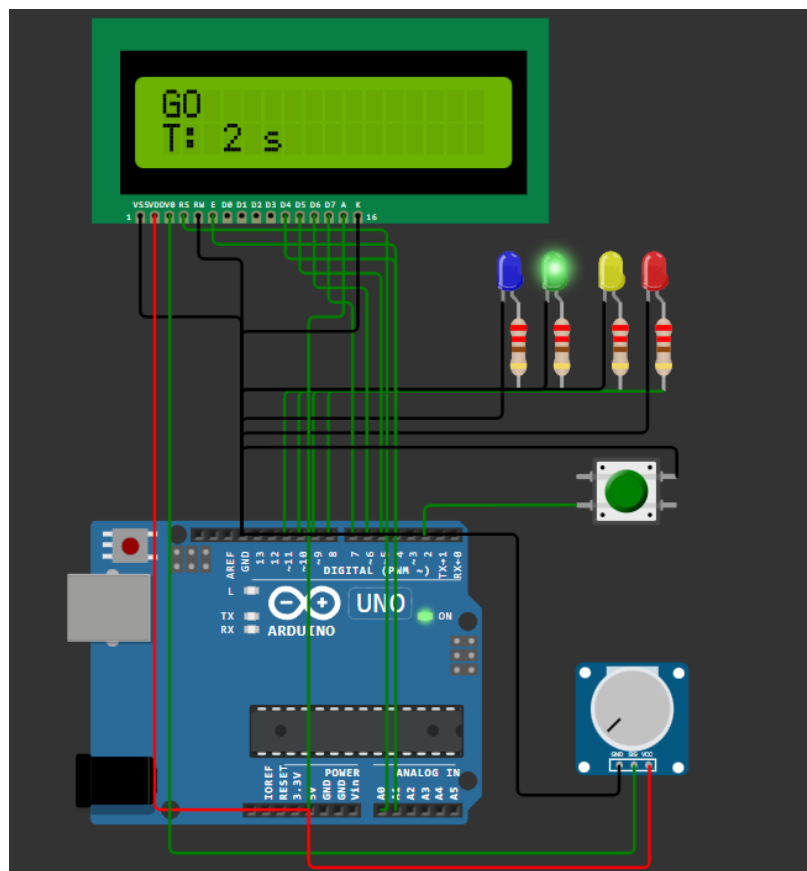


Question 5 — Combined Challenge: Traffic-Light Controller with Pedestrian Crossing



This project simulates a traffic light system using Arduino Uno, four LEDs, one push button, and a 16x2 LCD. The red, yellow, green, and pedestrian LEDs are connected to pins D8, D9, D10, and D11 through 220-ohm resistors, while all LED cathodes are connected to GND. The pedestrian push button is connected to D2 and GND and uses the internal pull-up resistor with external interrupt INT0. The LCD works in 4-bit mode where RS is connected to A0, EN to A1, and data pins D4–D7 are connected to Arduino pins D4–D7. The LCD VSS and K pins are connected to GND, VDD and A are connected to 5V, and the potentiometer is connected to V0 to control the screen contrast. The program changes automatically between GREEN, YELLOW, and RED states using timers, and when the pedestrian button is pressed during the green state, the system switches to pedestrian crossing mode where the pedestrian LED blinks and the LCD displays the current state and remaining time.